

Physics

Date: 30/07/23

PAGE No.:

DATE: / /

① Convex lens

② $P \propto \frac{1}{f}$

$\therefore$ focal length of 10 cm has greater power

③ Given

$$u = -5 \text{ cm}$$

$$v = -10 \text{ cm} \quad (\text{Behind the lens means real Image is formed})$$

By lens formula.

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{f} = \frac{1}{-10} - \frac{1}{-5} \Rightarrow \frac{-1}{10} + \frac{1}{5}$$

$$\Rightarrow \frac{-1 + 2}{10} = \frac{1}{10} = \frac{1}{f}$$

$$\boxed{f = +10 \text{ cm}}$$

The type of lens used is Convex lens as real & inverted image is formed and focal length is positive.

④ Given

$$v = +50 \text{ cm}$$

$m = -1$ (Real & inverted & same size image)

$$m = \frac{v}{u}$$

$$-1 = \frac{50}{u}$$

$$u = \frac{50}{-1} \Rightarrow u = -50 \text{ cm}$$

By Lens formula,

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{f} = \frac{1}{50} - \frac{1}{-50} = \frac{1}{50} + \frac{1}{50} = \frac{2}{50}$$

$$\frac{1}{f} = \frac{1}{25} \Rightarrow \boxed{f = 25 \text{ cm}}$$

$$\therefore \text{Power} \Rightarrow P = \frac{1}{f}$$

$$P = \frac{1}{25} \times 100 \text{ D} = \boxed{4 \text{ D}}$$